

LAB 6

NAME: Hajra Sarwar

REG NO: 2023-BSE-022

CLASS: BSE-5A

SUBJECT: CC

LAB TITLE: Linux Users, Groups, Permissions, Pipes, and Bash Scripting

TASK:

Task 1: Switch to root with su – and back to a normal user

Goal:

Demonstrate switching to the root account using su – and then returning to the normal user.

Steps Performed:

1. **Set a root password** (Ubuntu root is disabled by default; this step enables su – temporarily for the lab):

sudo passwd root

```
hajra@ubuntu-lab:~$ sudo passwd root
[sudo] password for hajra:
New password:
Retype new password:
passwd: password updated successfully
```

- o Entered a temporary root password for this lab.

2. **Switch to the root user and verify:**

su –

```
hajra@ubuntu-lab:~$ su -
Password:
root@ubuntu-lab:~#
```

Whoami

```
root@ubuntu-lab:~# whoami
root
```

Id

```
root@ubuntu-lab:~# id
uid=0(root) gid=0(root) groups=0(root)
```

- o Confirmed that the current user is root.

3. **Switch back to the normal user and verify:**

Exit

```
root@ubuntu-lab:~# exit
logout
hajra@ubuntu-lab:~$
```

Whoami

```
hajra@ubuntu-lab:~$ whoami
hajra
```

- o Returned to the normal user successfully.
-

Task 2: Create user *tom* and verify in passwd/group/shadow

Goal:

Create a user named tom, then verify the account details in the system files.

Steps Performed:

1. Create a new user tom (interactive command that sets a password and creates a home directory):

sudo adduser tom

```
hajra@ubuntu-lab:~$ sudo adduser tom
[sudo] password for hajra:
info: Adding user `tom' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `tom' (1002) ...
info: Adding new user `tom' (1002) with group `tom' (1002) ...
info: Creating home directory `/home/tom' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for tom
Enter the new value, or press ENTER for the default
  Full Name []: tom ch
    Room Number []: 2023-BSE-022
    Work Phone []: 0300-0000000
    Home Phone []: 0313-4567890
    Other []: Software Engineer
Is the information correct? [y/n] Y
info: Adding new user `tom' to supplemental / extra groups `users' ...
info: Adding user `tom' to group `users' ...
hajra@ubuntu-lab:~$
```

- o Entered password and user information as prompted.

2. Verify user tom in the /etc/passwd file:

cat /etc/passwd

```
hajra@ubuntu-lab:~$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mail List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534:/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
systemd-timesync:x:997:997:systemd Time Synchronization:/:/usr/sbin/nologin
dhcpcd:x:100:65534:DHCP Client Daemon,,,:/usr/lib/dhcpcd:/bin/false
messagingbus:x:101:102:/nonexistent:/usr/sbin/nologin
systemd-resolve:x:992:992:systemd Resolver:/:/usr/sbin/nologin
pollinate:x:102:1:/var/cache/pollinate:/bin/false
polkitd:x:991:991:User for polkitd:/usr/sbin/nologin
syslog:x:103:104:/nonexistent:/usr/sbin/nologin
uuidd:x:104:105:/run/uuidd:/usr/sbin/nologin
tcpdump:x:105:107:/nonexistent:/usr/sbin/nologin
tss:x:106:100:100:software stack,,,:/var/lib/tss:/bin/false
landscape:x:107:109:/var/lib/landscape:/usr/sbin/nologin
fwupd-refresh:x:989:989:Firmware update daemon:/var/lib/fwupd:/usr/sbin/nologin
usbmux:x:108:46:usbmux daemon,,,:/var/lib/usbmux:/usr/sbin/nologin
shd:x:109:65534:/run/shd:/usr/sbin/nologin
hajra:x:1000:1000:ubuntu:/home/hajra:/bin/bash
lxd:x:999:101:/var/snap/lxd/common/lxd:/bin/false
rtkit:x:110:110:RealtimeKit,,,:/proc:/usr/sbin/nologin
dnsmasq:x:996:65534:dnsmasq:/var/lib/isc:/usr/sbin/nologin
cups-pk-helper:x:111:111:user for cups-pk-helper service,,,:/nonexistent:/usr/sbin/nologin
lightdm:x:112:113:Light Display Manager:/var/lib/lightdm:/bin/false
whoopsie:x:113:115:/nonexistent:/bin/false
avahi:x:114:118:Avahi mDNS daemon,,,:/run/avahi-daemon:/usr/sbin/nologin
saned:x:115:120:/var/lib/saned:/usr/sbin/nologin
colord:x:116:121:colord colour management daemon,,,:/var/lib/colord:/usr/sbin/nologin
pulse:x:117:122:PulseAudio daemon,,,:/run/pulse:/usr/sbin/nologin
cups-browsed:x:118:111:/nonexistent:/usr/sbin/nologin
xrdp:x:119:124:/run/xrdp:/usr/sbin/nologin
tom:x:1002:1002:tom ch,2023-BSE-022,0300-0000000,0313-4567890,Software Engineer:/home/tom:/bin/bash
```

- o Confirmed that a new entry for tom exists, indicating the user account and home directory.

3. Verify user tom in the /etc/group file:

cat /etc/group

```
hajra@ubuntu-lab:~$ cat /etc/group
root:x:0:
daemon:x:1:
bin:x:2:
sys:x:3:
adm:x:4:syslog,hajra
tty:x:5:
disk:x:6:
pulse:x:122:
pulse-access:x:123:
xrdp:x:124:
docker:x:988:hajra
tom:x:1002:
hajra@ubuntu-lab:~$
```

- o Verified that tom is listed in the appropriate user group.

4. Verify user tom in the /etc/shadow file:

sudo cat /etc/shadow

```
xrdp:!:20385: : : :
tom:$y$j9T$ZLakmQ1Zn2tLKSuHEbbuZ0$L5F3YkskEfPNXwf/kh3YBhUM4JpoTKIkeSuEy9UM.n1:20392:0:99999:7: : :
hajra@ubuntu-lab:~$
```

- o Confirmed that tom has an entry with a hashed password (not visible in plaintext).

Task 3: Create groups; change tom's primary and secondary groups

Goal:

Create groups developer, devops, and designer. Then change tom's primary group and manage secondary group memberships.

Steps Performed:

1. Create the required groups and verify:

```
sudo groupadd developer
sudo groupadd devops
sudo groupadd designer
```

```
hajra@ubuntu-lab:~$ sudo groupadd developer
[sudo] password for hajra:
hajra@ubuntu-lab:~$ sudo groupadd devops
hajra@ubuntu-lab:~$ sudo groupadd designer
```

```
cat /etc/group
```

```
hajra@ubuntu-lab:~$ cat /etc/group
root:x:0:
daemon:x:1:
bin:x:2:
sys:x:3:
tom:x:1002:
developer:x:1003:
devops:x:1004:
designer:x:1005:
hajra@ubuntu-lab:~$
```

- Verified that the new groups developer, devops, and designer were successfully added in /etc/group.

2. Change tom's primary group to designer and verify:

```
sudo usermod -g designer tom
id tom
```

```
hajra@ubuntu-lab:~$ sudo usermod -g designer tom
[sudo] password for hajra:
hajra@ubuntu-lab:~$ id tom
uid=1002(tom) gid=1005(designer) groups=1005(designer),100(users)
```

- Confirmed that designer is now set as tom's primary group.

3. Add secondary groups developer and devops to tom and verify:

```
sudo usermod -aG developer,devops tom
id tom
```

```
groups tom
```

```
hajra@ubuntu-lab:~$ sudo usermod -aG developer,devops tom
hajra@ubuntu-lab:~$ id tom
uid=1002(tom) gid=1005(designer) groups=1005(designer),100(users),1003(developer),1004(devops)
hajra@ubuntu-lab:~$ groups tom
tom : designer users developer devops
hajra@ubuntu-lab:~$
```

- Verified that tom now belongs to the developer and devops groups as secondary groups.

4. Reset secondary groups so only tom (user's own group) remains and verify:

```
sudo usermod -G tom tom
id tom
groups tom
```

```
hajra@ubuntu-lab:~$ sudo usermod -G tom tom
hajra@ubuntu-lab:~$ id tom
uid=1002(tom) gid=1005(designer) groups=1005(designer),1002(tom)
hajra@ubuntu-lab:~$ groups tom
tom : designer tom
hajra@ubuntu-lab:~$
```

- Confirmed that all additional secondary groups were removed, leaving only tom's default group.

Task 4: Create/Delete Users (Jerry, Scooby) and Groups (jolly, anime)

Goal:

Create users using both adduser and useradd, demonstrate login, password, and home directory differences, then delete the created users and groups.

Steps Performed:

1. Create new users Jerry and Scooby:

sudo adduser Jerry

sudo useradd Scooby

```
hajra@ubuntu-lab:~$ sudo adduser jerry
info: Adding user `jerry' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `jerry' (1006) ...
info: Adding new user `jerry' (1006) with group `jerry' (1006) ...
info: Creating home directory `/home/jerry' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for jerry
Enter the new value, or press ENTER for the default
  Full Name []: jerry ch
    Room Number []: 15
    Work Phone []: 0300-0000000
    Home Phone []: 0300-1345678
    Other []: SE
Is the information correct? [Y/n] Y
info: Adding new user `jerry' to supplemental / extra groups `users' ...
info: Adding user `jerry' to group `users' ...
hajra@ubuntu-lab:~$ sudo useradd Scooby
hajra@ubuntu-lab:~$
```

- adduser automatically created a home directory and prompted for a password.
- useradd created a basic user account without a home directory or password.

2. Attempt to log in as Scooby before setting a password (expected failure):

su - Scooby

```
hajra@ubuntu-lab:~$ su - Scooby
Password:
su: Authentication failure
hajra@ubuntu-lab:~$
```

- Authentication failed because Scooby had no password.

3. Set a password for Scooby:

sudo passwd Scooby

```
hajra@ubuntu-lab:~$ sudo passwd Scooby
New password:
Retype new password:
passwd: password updated successfully
hajra@ubuntu-lab:~$
```

- Assigned a password successfully.

4. Log in again as Scooby (observe missing home directory warning):

su - Scooby

```
hajra@ubuntu-lab:~$ su - Scooby
Password:
su: warning: cannot change directory to /home/Scooby: No such file or directory
$
```

- System displayed: “No directory, logging in with HOME=/”.

5. Verify Scooby’s home directory and /etc/passwd entry:

Exit

cat /etc/passwd

```
$ exit
hajra@ubuntu-lab:~$ cat /etc/passwd
jerry:x:1006:1006:jerry ch,15,0300-0000000,0300-1345678,SE:/home/jerry:/bin/bash
Scooby:x:1007:1007::/home/Scooby:/bin/sh
```

ls -ld /home/Scooby

```
hajra@ubuntu-lab:~$ ls -ld /home/Scooby
ls: cannot access '/home/Scooby': No such file or directory
hajra@ubuntu-lab:~$
```

- Confirmed that /home/Scooby did not exist.

6. **Manually create Scooby's home directory and set ownership/permissions:**

```
sudo mkdir -p /home/Scooby
sudo chown Scooby:Scooby /home/Scooby
sudo chmod 750 /home/Scooby
ls -ld /home/Scooby
hajra@ubuntu-lab:~$ sudo mkdir -p /home/Scooby
hajra@ubuntu-lab:~$ sudo chown Scooby:Scooby /home/Scooby
hajra@ubuntu-lab:~$ sudo chmod 750 /home/Scooby
hajra@ubuntu-lab:~$ ls -ld /home/Scooby
drwxr-x--- 2 Scooby Scooby 4096 Nov  1 06:39 /home/Scooby
hajra@ubuntu-lab:~$
```

- o Successfully created and secured the directory.

7. **Log in as Scooby again to confirm correct home directory access:**

```
su - Scooby
pwd
ls -la
hajra@ubuntu-lab:~$ su - Scooby
Password:
$ pwd
/home/Scooby
$ ls -la
total 8
drwxr-x--- 2 Scooby Scooby 4096 Nov  1 06:39 .
drwxr-xr-x 6 root    root   4096 Nov  1 06:39 ..
$
```

- o Verified proper login into /home/Scooby.

8. **Verify users in /etc/passwd and check Scooby's shell:**

```
exit
cat /etc/passwd
$ exit
hajra@ubuntu-lab:~$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
jerry:x:1006:1006:jerry ch,15,0300-0000000,0300-1345678,SE:/home/jerry:/bin/bash
Scooby:x:1007:1007:/home/Scooby:/bin/sh
hajra@ubuntu-lab:~$
```

- o Observed Scooby's shell set to /bin/sh.

9. **Change Scooby's shell from /bin/sh to /bin/bash and verify:**

```
sudo usermod -s /bin/bash Scooby
su - Scooby
hajra@ubuntu-lab:~$ sudo usermod -s /bin/bash Scooby
hajra@ubuntu-lab:~$ su - Scooby
Password:
Scooby@ubuntu-lab:~$
```

- o Verified new default shell as /bin/bash.

10. **Create groups jolly and anime:**

```
sudo addgroup jolly
sudo groupadd anime
Scooby@ubuntu-lab:~$ exit
logout
hajra@ubuntu-lab:~$ sudo addgroup jolly
info: Selecting GID from range 1000 to 59999 ...
info: Adding group `jolly' (GID 1008) ...
hajra@ubuntu-lab:~$ sudo groupadd anime
hajra@ubuntu-lab:~$
```

11. **Verify groups in /etc/group:**

```
cat /etc/group
hajra@ubuntu-lab:~$ cat /etc/group
root:x:0:
daemon:x:1:
jolly:x:1008:
anime:x:1009:
hajra@ubuntu-lab:~$
```

- o Confirmed successful creation of both groups.

12. **Delete the groups and users:**

```
sudo delgroup jolly
sudo groupdel anime
```

```
cat /etc/group
```

```
hajra@ubuntu-lab:~$ sudo delgroup jolly
info: Removing group `jolly' ...
hajra@ubuntu-lab:~$ sudo groupdel anime
hajra@ubuntu-lab:~$ cat /etc/group
root:x:0:
daemon:x:1:
developer:x:1003:
devops:x:1004:
designer:x:1005:
jerry:x:1006:
Scooby:x:1007:
hajra@ubuntu-lab:~$
```

```
sudo deluser --remove-home Jerry
```

```
sudo userdel -r Scooby
```

```
cat /etc/passwd
```

```
hajra@ubuntu-lab:~$ sudo deluser --remove-home jerry
info: Looking for files to backup/remove ...
info: Removing files ...
info: Removing crontab ...
info: Removing user `jerry' ...
hajra@ubuntu-lab:~$ sudo userdel -r Scooby
userdel: user `Scooby' does not exist
hajra@ubuntu-lab:~$ cat /etc/passwd
root:x:0:root:/root:/bin/bash
daemon:x:1:daemon:/usr/sbin:/usr/sbin/nologin
cups-browsed:x:118:111::/nonexistent:/usr/sbin/nologin
xrdp:x:119:124::/run/xrdp:/usr/sbin/nologin
tom:x:1002:1005:tom ch,2023-BSE-022,0300-0000000,0313-4567890,Software Engineer:/home/tom:/bin/bash
hajra@ubuntu-lab:~$
```

- Verified that both users and groups were completely removed.

Task 5: Create User “Student”; Create Files; Set Owner/Group; Identify File Types

Goal:

Create a new user named Student, create files and directories, change ownership and group, and identify different file types including system special files.

Steps Performed:

1. Create a new user Student:

```
sudo adduser Student
```

```
hajra@ubuntu-lab:~$ sudo adduser student
info: Adding user `student' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `student' (1006) ...
info: Adding new user `student' (1006) with group `student (1006)' ...
info: Creating home directory `/home/student' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for student
Enter the new value, or press ENTER for the default
  Full Name []: student1
    Room Number []: 14
    Work Phone []: 0300-0000000
    Home Phone []: 0313-1345678
    Other []: SE
Is the information correct? [Y/n] Y
info: Adding new user `student' to supplemental / extra groups `users' ...
info: Adding user `student' to group `users' ...
hajra@ubuntu-lab:~$
```

- Entered a password and completed the setup for the Student user.

2. Switch to Student and create files and directories:

```
su - Student
```

```
touch file1
```

```
mkdir -p dir1
```

```
touch dir1/file2
```

```
ls -l
```

```

hajra@ubuntu-lab:~$ su - student
Password:
student@ubuntu-lab:~$ touch file1
student@ubuntu-lab:~$ mkdir -p dir1
student@ubuntu-lab:~$ touch dir1/file2
student@ubuntu-lab:~$ ls -l
total 4
drwxrwxr-x 2 student student 4096 Nov  1 08:12 dir1
-rw-rw-r-- 1 student student  0 Nov  1 08:10 file1
student@ubuntu-lab:~$

```

- Created a file file1, a directory dir1, and a second file file2 inside it.

3. Change the owner of file1 to tom and verify:

sudo chown tom file1

ls -l file1

```

hajra@ubuntu-lab:~$ sudo chown tom /home/student/file1
hajra@ubuntu-lab:~$ sudo ls -l /home/student/file1
-rw-rw-r-- 1 tom student 0 Nov  1 08:10 /home/student/file1
hajra@ubuntu-lab:~$

```

- Confirmed that the file owner was updated to tom.

4. Change the group of file1 to devops and verify:

sudo chgrp devops file1

ls -l file1

```

hajra@ubuntu-lab:~$ sudo chgrp devops /home/student/file1
hajra@ubuntu-lab:~$ sudo ls -l /home/student/file1
-rw-rw-r-- 1 tom devops 0 Nov  1 08:10 /home/student/file1
hajra@ubuntu-lab:~$

```

- Verified that the file's group ownership changed to devops.

5. Identify files, directories, and special files:

ls -l

ls -l dir1

ls -l /dev/null

file file1 dir1 /dev/null

```

hajra@ubuntu-lab:~$ su - student
Password:
student@ubuntu-lab:~$ ls -l
total 4
drwxrwxr-x 2 student student 4096 Nov  1 08:12 dir1
-rw-rw-r-- 1 tom devops  0 Nov  1 08:10 file1
student@ubuntu-lab:~$ ls -l dir1
total 0
-rw-rw-r-- 1 student student 0 Nov  1 08:12 file2
student@ubuntu-lab:~$ ls -l /dev/null
crw-rw-rw- 1 root root 1, 3 Nov  4 07:07 /dev/null
student@ubuntu-lab:~$ file file1 dir1 /dev/null
file1:      empty
dir1:       directory
/dev/null:  character special (1/3)
student@ubuntu-lab:~$

```

- Observed file types:
 - file1 → regular file
 - dir1 → directory
 - /dev/null → special device file

6. Exit from Student account:

Exit

```

student@ubuntu-lab:~$ exit
logout
hajra@ubuntu-lab:~$

```

- Returned to the previous user successfully.

Task 6: Change Permissions Using Symbolic Mode

Goal:

Modify file permissions for ~/file1 using symbolic notation (r, w, x, u, g, o, and a) to demonstrate adding and removing different access levels.

Steps Performed:

1. **Ensure the Student user and target file exist:**

```
su - Student
cd ~
ls -l file1
hajra@ubuntu-lab:~$ su - student
Password:
student@ubuntu-lab:~$ cd ~
student@ubuntu-lab:~$ ls -l file1
-rw-rw-r-- 1 tom devops 0 Nov  1 08:10 file1
student@ubuntu-lab:~$
```

- Confirmed presence of file1 in the home directory.

2. **Remove all permissions from file1:**

```
chmod -rwx file1
ls -l file1
hajra@ubuntu-lab:~$ sudo chown student /home/student/file1
[sudo] password for hajra:
hajra@ubuntu-lab:~$ su - student
Password:
student@ubuntu-lab:~$ chmod -rwx file1
student@ubuntu-lab:~$ ls -l file1
----- 1 student devops 0 Nov  1 08:10 file1
student@ubuntu-lab:~$
```

- Verified that all permissions were removed (-----).

3. **Add read permission to all users:**

```
chmod +r file1
ls -l file1
student@ubuntu-lab:~$ chmod +r file1
student@ubuntu-lab:~$ ls -l file1
-r--r--r-- 1 student devops 0 Nov  1 08:10 file1
student@ubuntu-lab:~$
```

- Confirmed that the file now has read (r) permission for user, group, and others.

4. **Add execute permission to the user (owner):**

```
chmod u+x file1
ls -l file1
student@ubuntu-lab:~$ chmod u+x file1
student@ubuntu-lab:~$ ls -l file1
-r-xr--r-- 1 student devops 0 Nov  1 08:10 file1
student@ubuntu-lab:~$
```

- Verified that the user now has execute (x) permission.

5. **Add write permission to both user and group:**

```
chmod ug+w file1
ls -l file1
student@ubuntu-lab:~$ chmod ug+w file1
student@ubuntu-lab:~$ ls -l file1
-rwxrw-r-- 1 student devops 0 Nov  1 08:10 file1
student@ubuntu-lab:~$
```

- Confirmed that both user and group gained write (w) permission.

6. **Remove all permissions explicitly from user, group, and others:**

```
chmod ugo-rwx file1
ls -l file1
student@ubuntu-lab:~$ chmod ugo-rwx file1
student@ubuntu-lab:~$ ls -l file1
----- 1 student devops 0 Nov  1 08:10 file1
student@ubuntu-lab:~$
```

- Verified that no permissions remain (-----).

Task 7: Change Permissions Using “Set” Symbolic Form (u= g= o=)

Goal:

Use the “set” symbolic form (u=, g=, o=) to assign and remove specific file permissions for the user, group, and others.

Steps Performed:

1. **Ensure you are logged in as Student and confirm the file exists:**

```
su - Student
cd ~
ls -l file1
hajra@ubuntu-lab:~$ su - student
Password:
student@ubuntu-lab:~$ cd ~
student@ubuntu-lab:~$ ls -l file1
----- 1 student devops 0 Nov  1 08:10 file1
student@ubuntu-lab:~$
```

- Verified that the current user is Student and the target file file1 exists in the home directory.

2. **Set all permissions (read, write, execute) for user, group, and others:**

```
chmod u=rwx,g=rwx,o=rwx file1
ls -l file1
student@ubuntu-lab:~$ chmod u=rwx,g=rwx,o=rwx file1
student@ubuntu-lab:~$ ls -l file1
-rwxrwxrwx 1 student devops 0 Nov  1 08:10 file1
student@ubuntu-lab:~$
```

- Confirmed that all permissions (rwxrwxrwx) were applied to the file.

3. **Remove execute permission from group and others:**

```
chmod g=rw,o=rw file1
ls -l file1
student@ubuntu-lab:~$ chmod g=rw,o=rw file1
student@ubuntu-lab:~$ ls -l file1
-rwxrw-rw- 1 student devops 0 Nov  1 08:10 file1
student@ubuntu-lab:~$
```

- Verified that execute permission (x) was removed for both group and others, leaving (rwxrw-rw-).

4. **Remove all permissions from user, group, and others:**

```
chmod u=,g=,o= file1
ls -l file1
student@ubuntu-lab:~$ chmod u=,g=,o= file1
student@ubuntu-lab:~$ ls -l file1
----- 1 student devops 0 Nov  1 08:10 file1
student@ubuntu-lab:~$
```

- Confirmed that no permissions remain (-----).

Task 8: Change Permissions Using Numeric (Octal) Mode

Goal:

Demonstrate how to modify file permissions using **numeric (octal)** values to represent different combinations of read (r), write (w), and execute (x) permissions.

Steps Performed:

1. **Ensure you are logged in as Student and the file exists:**

```
su - Student
```

```
cd ~
ls -l file1
hajra@ubuntu-lab:~$ su - student
Password:
student@ubuntu-lab:~$ cd ~
student@ubuntu-lab:~$ ls -l file1
----- 1 student devops 0 Nov  1 08:10 file1
```

- Verified that the current user is Student and file1 exists in the home directory.

2. **Set permissions to 777 (rwx for user, group, and others):**

```
chmod 777 file1
ls -l file1
student@ubuntu-lab:~$ chmod 777 file1
student@ubuntu-lab:~$ ls -l file1
-rwxrwxrwx 1 student devops 0 Nov  1 08:10 file1
```

- Verified full permissions: rwxrwxrwx

3. **Set permissions to 700 (rwx for user only):**

```
chmod 700 file1
ls -l file1
student@ubuntu-lab:~$ chmod 700 file1
student@ubuntu-lab:~$ ls -l file1
-rwx----- 1 student devops 0 Nov  1 08:10 file1
```

- Verified permissions: rwx-----

4. **Set permissions to 744 (rwx for user, read for group and others):**

```
chmod 744 file1
ls -l file1
student@ubuntu-lab:~$ chmod 744 file1
student@ubuntu-lab:~$ ls -l file1
-rwxr--r-- 1 student devops 0 Nov  1 08:10 file1
```

- Verified permissions: rwxr--r--

5. **Set permissions to 640 (rw for user, r for group, none for others):**

```
chmod 640 file1
ls -l file1
student@ubuntu-lab:~$ chmod 640 file1
student@ubuntu-lab:~$ ls -l file1
-rw-r----- 1 student devops 0 Nov  1 08:10 file1
```

- Verified permissions: rw-r-----

6. **Set permissions to 664 (rw for user and group, read for others):**

```
chmod 664 file1
ls -l file1
student@ubuntu-lab:~$ chmod 644 file1
student@ubuntu-lab:~$ ls -l file1
-rw-r--r-- 1 student devops 0 Nov  1 08:10 file1
```

- Verified permissions: rw-rw-r--

7. **Set permissions to 775 (rwx for user and group, read/execute for others):**

```
chmod 775 file1
ls -l file1
student@ubuntu-lab:~$ chmod 755 file1
student@ubuntu-lab:~$ ls -l file1
-rwxr-xr-x 1 student devops 0 Nov  1 08:10 file1
```

- Verified permissions: rwxrwxr-x

8. **Set permissions to 750 (rwx for user, read/execute for group, none for others):**

```
chmod 750 file1
ls -l file1
student@ubuntu-lab:~$ chmod 750 file1
student@ubuntu-lab:~$ ls -l file1
-rwxr-x--- 1 student devops 0 Nov  1 08:10 file1
```

- Verified permissions: rwxr-x---
-

```
sudo grep -E 'fail|error' /var/log/syslog | head
```


4. Redirect search results to a new file (overwrite mode):

```
sudo grep -i systemd /var/log/syslog > ~/syslog_systemd.txt
```

```
hajra@ubuntu-lab: $ sudo grep -i systemd /var/log/syslog > ~/syslog_systemd.txt
grep: /var/log/syslog: binary file matches
hajra@ubuntu-lab: $
```

- Created (or replaced) a file named `syslog_systemd.txt` containing all “systemd” log entries.

5. Append additional search results to the same file:

```
sudo grep -i network /var/log/syslog >> ~/syslog_systemd.txt
```

```
cat ~/syslog_systemd.txt
```

```

nmapgrabubuntu-lab:~$ sudo grep -i network /var/log/syslog >> -syslog_systemd.txt
grep: /var/log/syslog: binary file matches
nmapgrabubuntu-lab:~$ cat -syslog_systemd.txt
2025-10-31T04:45:03.690810+00:00 ubuntu-lab systemd[1]: rsyslog.service: Sent signal SIGUSR1 to main process 1555 (rsyslogd) on client request.
2025-10-31T04:45:03.739660+00:00 ubuntu-lab systemd[1]: logrotate.service: Deactivated successfully.
2025-10-31T04:45:03.739660+00:00 ubuntu-lab systemd[1]: Finished logrotate.service - Rotate log files.
2025-10-31T04:45:04.320004+00:00 ubuntu-lab systemd[1]: snap.rootfs.70mfc.mount: Deactivated successfully.
2025-10-31T04:45:04.320004+00:00 ubuntu-lab systemd[1]: Started containerd.service - containerd container runtime.
2025-10-31T04:45:05.174257+00:00 ubuntu-lab systemd[1]: Starting docker.service - Docker Application Container Engine...
2025-10-31T04:45:05.174519+00:00 ubuntu-lab systemd[1]: dmesg.service: Deactivated successfully.
2025-10-31T04:45:05.186492+00:00 ubuntu-lab systemd[1]: Cannot find unit for notify message of PID 2250, ignoring.
2025-10-31T04:45:05.206909+00:00 ubuntu-lab systemd[1]: snap.rootfs.2069m.mount: Deactivated successfully.
2025-10-31T04:45:05.214210+00:00 ubuntu-lab systemd[1]: snap.rootfs.LW0N093.mount: Deactivated successfully.
2025-10-31T04:45:05.286891+00:00 ubuntu-lab systemd[1]: snap.microk8s.daemon-etc.service: Deactivated successfully.
2025-10-31T04:45:05.287358+00:00 ubuntu-lab systemd[1]: snap.microk8s.daemon-flanneld.service: Deactivated successfully.
2025-10-31T04:45:05.287358+00:00 ubuntu-lab systemd[1]: snap.microk8s.daemon-apiserver-proxy.service: Deactivated successfully.
2025-10-31T04:45:05.452838+00:00 ubuntu-lab systemd[1]: Flatpak crash report service - automatic crash report generation.
2025-10-31T04:45:05.539552+00:00 ubuntu-lab systemd[1]: snap.syscheck.v2xdmountpoint.v2xd3413822832.mount: Deactivated successfully.
2025-10-31T04:45:06.000919+00:00 ubuntu-lab systemd[1]: NetworkManager-dispatcher.service: Deactivated successfully.
2025-10-31T04:45:06.124799+00:00 ubuntu-lab systemd[1]: snap.microk8s.daemon-containerd.service: Got notification message from PID 2308, but reception only permitted for main PID 1721
2025-10-31T04:45:06.269929+00:00 ubuntu-lab systemd[1]: Cannot find unit for notify message of PID 2310, ignoring.
2025-10-31T04:45:06.269929+00:00 ubuntu-lab systemd[1]: Reloading apparmor.service - Load AppArmor profiles...
2025-10-31T04:45:06.270865+00:00 ubuntu-lab systemd[1]: snap.microk8s.daemon-containerd.service: Got notification message from PID 2297, but reception only permitted for main PID 1721
2025-10-31T04:45:06.411841+00:00 ubuntu-lab apparmor.service[2314]: Restarting AppArmor
2025-10-31T04:45:06.489552+00:00 ubuntu-lab apparmor.service[2314]: /lib/apparmor/apparmor.systemd: 148: [: Illegal number: yes
2025-10-31T04:45:06.489552+00:00 ubuntu-lab apparmor.service[2314]: Reloading AppArmor profiles
2025-10-31T04:45:06.908014+00:00 ubuntu-lab systemd[1]: Started snapd.service - Snap Daemon.
2025-10-31T04:45:06.916824+00:00 ubuntu-lab systemd[1]: Finished snapd.seeded.service - Wait until snapd is fully seeded.
2025-10-31T04:45:07.010004+00:00 ubuntu-lab systemd[1]: snapd.autoupdate.service - Auto Import assertions from block devices was skipped because no trigger condition checks were met.
2025-10-31T04:45:07.011558+00:00 ubuntu-lab dbus-daemon[1043]: [system] Activating via systemd: service name='org.freedesktop.timedate1' unit='dbus-org.freedesktop.timedate1.service' requested by '11.25' (uid=0 pid=1449 comm="/snap/snapd/current/usr/lib/snap/snapd" label="unconfined")
2025-10-31T04:45:07.040325+00:00 ubuntu-lab systemd[1]: Starting systemd-timedated.service - Time & Date Service...
2025-10-31T04:45:08.213012+00:00 ubuntu-lab systemd[1]: Started snap.keepalived.daemon.service - Service for snap application keepalived.daemon.
2025-10-31T04:45:09.412656+00:00 ubuntu-lab systemd[1]: Started systemd-timesyncd.service - System Time & Date Service.
2025-10-31T04:45:09.421565+00:00 ubuntu-lab dockerd[2249]: "2025-10-31T04:45:09.414268966Z" level=info msg="detected 127.0.0.53 nameserver, assuming systemd-resolved, so using resolv.c
onf: /run/systemd/resolve/resolv.conf"
2025-10-31T04:45:09.540799+00:00 ubuntu-lab systemd[1]: Reloaded apparmor.service - Load AppArmor profiles.
2025-10-31T04:45:09.712899+00:00 ubuntu-lab systemd[1]: snap.docker.7128m.mount: Deactivated successfully.
2025-10-31T04:45:09.730171+00:00 ubuntu-lab systemd[1]: Finished snap.docker.nvidia-container-toolkit.service - Service for snap application docker.nvidia-container-toolkit.
2025-10-31T04:45:09.731070+00:00 ubuntu-lab systemd[1]: snap.docker.nvidia-container-toolkit.service: Consumed 1.292s CPU time.
2025-10-31T04:45:09.726315+00:00 ubuntu-lab systemd[1]: Started snap.docker.dockerd.service - Service for snap application docker.dockerd.
2025-10-31T04:45:10.107426+00:00 ubuntu-lab systemd[1]: snap.lxd.activate.service: Deactivated successfully.
2025-10-31T04:45:10.132545+00:00 ubuntu-lab systemd[1]: Finished snap.lxd.activate.service - Service for snap application lxd.activate.
2025-10-31T04:45:10.113746+00:00 ubuntu-lab systemd[1]: snap.lxd.activate.service: Consumed 1.255s CPU time.

```

- Appended “network”-related entries to the same text file and verified the combined output.

6. (Alternative) Use `journalctl` to explore logs:

```
sudo journalctl | less
```

```

Oct 23 04:40:45 ubuntu-lab kernel: Linux version 6.8.0-85-generic (build@dclyc02-amd64-106) (x86_64-linux-gnu-gcc-13 (Ubuntu 13.3.0-6ubuntu2-24.04) 13.3.0, GNU ld (GNU Binutils for Ubuntu) 2.42) #85-Ubuntu SMP PREEMPT_DYNAMIC Thu Sep 15 12:56:59 UTC 2025 (Ubuntu 6.8.0-85-generic 6.8.12)
Oct 23 04:40:45 ubuntu-lab kernel: Command line: BOOT_IMAGE=/vmlinuz-6.8.0-85-generic root=/dev/mapper/ubuntu--vg-ubuntu--lv ro
Oct 23 04:40:45 ubuntu-lab kernel: KERNEL supported cpus:
Oct 23 04:40:45 ubuntu-lab kernel: Intel GenuineIntel
Oct 23 04:40:45 ubuntu-lab kernel: AMD AuthenticAMD
Oct 23 04:40:45 ubuntu-lab kernel: Hygon HygonGenuine
Oct 23 04:40:45 ubuntu-lab kernel: Centaur CentaurHauls
Oct 23 04:40:45 ubuntu-lab kernel: Zhaoxin Shanghai
Oct 23 04:40:45 ubuntu-lab kernel: Disabled fast string operations
Oct 23 04:40:45 ubuntu-lab kernel: BIOS-provided physical RAM map:
Oct 23 04:40:45 ubuntu-lab kernel: BIOS-e820: [mem 0x0000000000000000-0x000000000009c7ff] usable
Oct 23 04:40:45 ubuntu-lab kernel: BIOS-e820: [mem 0x000000000009e000-0x000000000000ffff] reserved
Oct 23 04:40:45 ubuntu-lab kernel: BIOS-e820: [mem 0x000000000000dc00-0x000000000000ffff] reserved
Oct 23 04:40:45 ubuntu-lab kernel: BIOS-e820: [mem 0x0000000000010000-0x000000000007edfff] usable
Oct 23 04:40:45 ubuntu-lab kernel: BIOS-e820: [mem 0x000000000007ee0000-0x000000000007efefff] ACPI data
Oct 23 04:40:45 ubuntu-lab kernel: BIOS-e820: [mem 0x000000000007eff000-0x000000000007fffff] ACPI NVS
Oct 23 04:40:45 ubuntu-lab kernel: BIOS-e820: [mem 0x000000000007ff0000-0x000000000007fffff] usable
Oct 23 04:40:45 ubuntu-lab kernel: BIOS-e820: [mem 0x0000000000000000-0x000000000007ffffff] reserved
Oct 23 04:40:45 ubuntu-lab kernel: BIOS-e820: [mem 0x0000000000ec0000-0x0000000000ec0fff] reserved
Oct 23 04:40:45 ubuntu-lab kernel: BIOS-e820: [mem 0x0000000000ec0000-0x0000000000ec0fff] reserved
Oct 23 04:40:45 ubuntu-lab kernel: BIOS-e820: [mem 0x0000000000ff0000-0x0000000000ff0fff] reserved
Oct 23 04:40:45 ubuntu-lab kernel: NX (Execute Disable) protection: active
Oct 23 04:40:45 ubuntu-lab kernel: AIC: Static calls initialized
Oct 23 04:40:45 ubuntu-lab kernel: SMIOs 2.7.2 present.
Oct 23 04:40:45 ubuntu-lab kernel: DMI: VMware, Inc. VMware Virtual Platform/4404BX Desktop Reference Platform, BIOS 6.00 11/12/2020
Oct 23 04:40:45 ubuntu-lab kernel: VMware: hypervisor model: 0x2
Oct 23 04:40:45 ubuntu-lab kernel: Hypervisor detected: VMware
Oct 23 04:40:45 ubuntu-lab kernel: VMware: TSC Freq read from hypervisor : 1895.997 MHz
Oct 23 04:40:45 ubuntu-lab kernel: VMware: Host bus clock speed read from hypervisor : 660000000 Hz
Oct 23 04:40:45 ubuntu-lab kernel: VMware: using clock offset of 15123608049 ns
Oct 23 04:40:45 ubuntu-lab kernel: tsc: Detected 1895.997 MHz processor
Oct 23 04:40:45 ubuntu-lab kernel: e820: update [mem 0x00000000-0x00000fff] usable ==> reserved
Oct 23 04:40:45 ubuntu-lab kernel: e820: remove [mem 0x0000a000-0x0000ffff] usable
Oct 23 04:40:45 ubuntu-lab kernel: init pfns = 0x0000 max_arch_pfn = 0x00000000
Oct 23 04:40:45 ubuntu-lab kernel: total RAM covered: 3072M
Oct 23 04:40:45 ubuntu-lab kernel: Found optimal setting for mtrr clean up
Oct 23 04:40:45 ubuntu-lab kernel: gran_size: 64k chunk_size: 64k num_reg: 2 lose cover RAM: 0G
Oct 23 04:40:45 ubuntu-lab kernel: MTRR map: 6 entries (5 fixed + 1 variable; max 21), built from 8 variable MTRRs
Oct 23 04:40:45 ubuntu-lab kernel: x86/PAT: Configuration [0-7]: WB UC UC- UC WB WB UC- WT
Oct 23 04:40:45 ubuntu-lab kernel: Found SMP MP-table at [mem 0x000f6a70-0x000f6a7f]
Oct 23 04:40:45 ubuntu-lab kernel: Using GPT pages for direct mapping
Oct 23 04:40:45 ubuntu-lab kernel: RAMDISK: RAMDISK: 0x2f5b5000-0x33ad1fff

```

```
sudo journalctl -u systemd | grep -i error > ~/journal_errors.txt
```

```
hajra@ubuntu-lab:~$ sudo journalctl -u systemd | grep -i error > ~/journal_errors.txt
```

```
hajra@ubuntu-lab:~$
```

- Viewed and filtered logs directly from systemd journal as an alternative to /var/log/syslog.

Task 10: Bash Script: setup.sh (Variables, Command Substitution, File/Dir Checks, Permissions)

Goal:

Create and run a Bash script named **setup.sh** using **vim**, adding code **incrementally** step-by-step. Each step was edited, saved, executed, and verified with screenshots of the vim editor and terminal output.

1. Add Bash Shebang

Code added:

#!/bin/bash

- Opened vim: vim setup.sh
- Inserted the shebang line, saved and quit (:wq).
- Made the script executable and ran it:
chmod +x setup.sh
./setup.sh

```
student@ubuntu-lab:~$ vim setup.sh
student@ubuntu-lab:~$ chmod +x setup.sh
student@ubuntu-lab:~$ ./setup.sh
student@ubuntu-lab:~$
```

2. Define and Display a Variable

```
student@ubuntu-lab: ~
#!/bin/bash
## Define and show var1
var1="Hello from Lab 6"
echo "var1: $var1"
```

- Edited with vim, appended the code, saved, and executed:
./setup.sh

```
student@ubuntu-lab:~$ ./setup.sh
var1: Hello from Lab 6
```

3. Command Substitution: Save and Display ls -l

```
# Save ls -l to variable and display
allFiles="$(ls -l)"
echo "allFiles (ls -l):"
echo "$allFiles"
```

- Appended in vim and executed the script again:
./setup.sh

```
student@ubuntu-lab:~$ ./setup.sh
var1: Hello from Lab 6
allFiles (ls -l):
total 8
drwxrwxr-x 2 student student 4096 Nov  1 08:12 dir1
-rwxr-x-- 1 student devops    0 Nov  1 08:10 file1
-rwxrwxr-x 1 student student 183 Nov  4 10:33 setup.sh
```

4. Directory Check and Creation

```
# Directory check
if [ -d "dir1" ]; then
    echo "Directory dir1 exists."
else
    echo "Directory dir1 does not exist. Creating..."
    mkdir -p "dir1"
    echo "Directory dir1 created."
fi
```

- Appended in vim and ran:
./setup.sh

```
student@ubuntu-lab:~$ ./setup.sh
var1: Hello from Lab 6
allFiles (ls -l):
total 8
drwxrwxr-x 2 student student 4096 Nov  1 08:12 dir1
-rwxr-x-- 1 student devops    0 Nov  1 08:10 file1
-rwxrwxr-x 1 student student 369 Nov  4 10:38 setup.sh
Directory dir1 exists.
```

5. File Check and Creation

```
# File check
if [ -f "dir1/file2" ]; then
    echo "file2 already exists."
else
    echo "file2 does not exist. Creating..."
    touch "dir1/file2"
    chmod a-rwx "dir1/file2"
    echo "file2 created."
fi
```

- Appended in vim and executed:

./setup.sh

```
student@ubuntu-lab: $ ./setup.sh
var1: Hello from Lab 6
allFiles (ls -l):
total 8
drwxrwxr-x 2 student student 4096 Nov  1 08:12 dir1
-rwxr-x-- 1 student devops    0 Nov  1 08:10 file1
-rwxrwxr-x 1 student student 567 Nov  4 10:57 setup.sh
Directory dir1 exists.
file2 already exists.
```

6. Permission Check and Adjustment

```
# Permission checks for dir1/file2 (user permissions)
fc "dir1/file2"
if [ ! -r "$f" ]; then
    echo "Read permission missing; granting to user..."
    chmod u+r "$f"
fi

if [ ! -w "$f" ]; then
    echo "Write permission missing; granting to user..."
    chmod u+w "$f"
fi

if [ ! -x "$f" ]; then
    echo "Execute permission missing; granting to user..."
    chmod u+x "$f"
fi

echo "Final permissions for $f:"
ls -l "$f"
```

- Appended this section in vim and ran:

./setup.sh

```
student@ubuntu-lab: $ ./setup.sh
var1: Hello from Lab 6
allFiles (ls -l):
total 8
drwxrwxr-x 2 student student 4096 Nov  1 08:12 dir1
-rwxr-x-- 1 student devops    0 Nov  1 08:10 file1
-rwxrwxr-x 1 student student 980 Nov  4 11:00 setup.sh
Directory dir1 exists.
file2 already exists.
Execute permission missing; granting to user...
Final permissions for dir1/file2:
-rwxr--r-- 1 student student 0 Nov  1 08:12 dir1/file2
```

Task 11: Script setup.sh – Argument Comparisons and String Checks

Goal:

Learn numeric and string comparison operators in Bash by adding one if-test at a time and running the script with example arguments.

Each step includes:

- Editing setup.sh in **vim** (show buffer screenshot)
- Saving and running the script with test arguments (show output screenshot)

1. Shebang, Variable Setup

```
hajra@ubuntu-lab: ~
#!/bin/bash
#num=$1
#str=$2
#
```

Commands:

```
chmod +x setup.sh  
./setup.sh 10 Student
```

```
hajra@ubuntu-lab:~$ chmod +x setup.sh  
hajra@ubuntu-lab:~$ ./setup.sh 10 student  
hajra@ubuntu-lab:~$
```

2. Numeric Equal (-eq)

```
hajra@ubuntu-lab:~$
```

```
#!/bin/bash  
num=$1  
str=$2  
  
if [ "$num" -eq 10 ]; then  
    echo "$num is equal to 10 (-eq)."  
else  
    echo "$num is NOT equal to 10 (-eq)."  
fi
```

Commands:

```
./setup.sh 10 Student  
./setup.sh 7 Student
```

```
hajra@ubuntu-lab:~$ ./setup.sh 10 Student  
10 is equal to 10 (-eq).  
hajra@ubuntu-lab:~$ ./setup.sh 7 Student  
7 is NOT equal to 10 (-eq).
```

3. Numeric Not Equal (-ne)

```
if [ "$num" -ne 10 ]; then  
    echo "$num is not equal to 10 (-ne)."  
else  
    echo "$num is equal to 10 (-ne false)."  
fi
```

Commands:

```
./setup.sh 7 Student  
./setup.sh 10 Student
```

```
hajra@ubuntu-lab:~$ ./setup.sh 7 Student  
7 is NOT equal to 10 (-eq).  
7 is not equal to 10 (-ne).  
hajra@ubuntu-lab:~$ ./setup.sh 10 Student  
10 is equal to 10 (-eq).  
10 is equal to 10 (-ne false).
```

4. Greater Than (-gt)

```
if [ "$num" -gt 10 ]; then  
    echo "$num is greater than 10 (-gt)."  
else  
    echo "$num is NOT greater than 10 (-gt)."  
fi
```

Commands:

```
./setup.sh 12 Student  
./setup.sh 9 Student
```

```
hajra@ubuntu-lab:~$ ./setup.sh 12 Student  
12 is NOT equal to 10 (-eq).  
12 is not equal to 10 (-ne).  
12 is greater than 10 (-gt).  
hajra@ubuntu-lab:~$ ./setup.sh 9 Student  
9 is NOT equal to 10 (-eq).  
9 is not equal to 10 (-ne).  
9 is NOT greater than 10 (-gt).
```

5. Less Than (-lt)

```
if [ "$num" -lt 10 ]; then  
    echo "$num is less than 10 (-lt)."  
else  
    echo "$num is NOT less than 10 (-lt)."  
fi
```

Commands:

./setup.sh 5 Student
./setup.sh 11 Student

```
hajra@ubuntu-lab:~$ ./setup.sh 5 Student
5 is NOT equal to 10 (-eq).
5 is not equal to 10 (-ne).
5 is NOT greater than 10 (-gt).
5 is less than 10 (-lt).
hajra@ubuntu-lab:~$ ./setup.sh 11 Student
11 is NOT equal to 10 (-eq).
11 is not equal to 10 (-ne).
11 is greater than 10 (-gt).
11 is NOT less than 10 (-lt).
```

6. Greater Than or Equal (-ge)

```
if [ "$num" -ge 10 ]; then
    echo "$num is greater than or equal to 10 (-ge)."
```

```
else
    echo "$num is NOT greater than or equal to 10 (-ge)."
```

```
fi
```

Commands:

./setup.sh 10 Student
./setup.sh 8 Student

```
hajra@ubuntu-lab:~$ ./setup.sh 10 Student
10 is equal to 10 (-eq).
10 is equal to 10 (-ne false).
10 is NOT greater than 10 (-gt).
10 is NOT less than 10 (-lt).
10 is greater than or equal to 10 (-ge).
hajra@ubuntu-lab:~$ ./setup.sh 8 Student
8 is NOT equal to 10 (-eq).
8 is not equal to 10 (-ne).
8 is NOT greater than 10 (-gt).
8 is less than 10 (-lt).
8 is NOT greater than or equal to 10 (-ge).
```

7. Less Than or Equal (-le)

```
if [ "$num" -le 10 ]; then
    echo "$num is less than or equal to 10 (-le)."
```

```
else
    echo "$num is NOT less than or equal to 10 (-le)."
```

```
fi
```

Commands:

./setup.sh 10 Student
./setup.sh 12 Student

```
hajra@ubuntu-lab:~$ ./setup.sh 10 Student
10 is equal to 10 (-eq).
10 is equal to 10 (-ne false).
10 is NOT greater than 10 (-gt).
10 is NOT less than 10 (-lt).
10 is greater than or equal to 10 (-ge).
10 is less than or equal to 10 (-le).
hajra@ubuntu-lab:~$ ./setup.sh 12 Student
12 is NOT equal to 10 (-eq).
12 is not equal to 10 (-ne).
12 is greater than 10 (-gt).
12 is NOT less than 10 (-lt).
12 is greater than or equal to 10 (-ge).
12 is NOT less than or equal to 10 (-le).
```

8. String Equality (=)

```
if [ "$str" = "Student" ]; then
    echo "Second argument equals 'Student' ( = )."
```

```
else
    echo "Second argument does NOT equal 'Student' ( = )."
```

```
fi
```

Commands:

./setup.sh 10 Student
./setup.sh 10 Test

```

raj@ubuntu-lab:~$ ./setup.sh 10 Student
10 is equal to 10 (-eq).
10 is equal to 10 (-ne false).
10 is NOT greater than 10 (-gt).
10 is NOT less than 10 (-lt).
10 is greater than or equal to 10 (-ge).
10 is less than or equal to 10 (-le).
Second argument equals 'Student' ( = ).
raj@ubuntu-lab:~$ ./setup.sh 10 Test
10 is equal to 10 (-eq).
10 is equal to 10 (-ne false).
10 is NOT greater than 10 (-gt).
10 is NOT less than 10 (-lt).
10 is greater than or equal to 10 (-ge).
10 is less than or equal to 10 (-le).
Second argument does NOT equal 'Student' ( != ).

```

9. String Inequality (!=)

```

if [ "$str" != "Student" ]; then
    echo "Second argument is not equal to 'Student' ( != )."
else
    echo "Second argument equals 'Student' ( != false)."

```

Commands:

`./setup.sh 10 Test`

`./setup.sh 10 Student`

```

raj@ubuntu-lab:~$ ./setup.sh 10 Test
10 is equal to 10 (-eq).
10 is equal to 10 (-ne false).
10 is NOT greater than 10 (-gt).
10 is NOT less than 10 (-lt).
10 is greater than or equal to 10 (-ge).
10 is less than or equal to 10 (-le).
Second argument does NOT equal 'Student' ( != ).
Second argument is not equal to 'Student' ( != ).
raj@ubuntu-lab:~$ ./setup.sh 10 Student
10 is equal to 10 (-eq).
10 is equal to 10 (-ne false).
10 is NOT greater than 10 (-gt).
10 is NOT less than 10 (-lt).
10 is greater than or equal to 10 (-ge).
10 is less than or equal to 10 (-le).
Second argument equals 'Student' ( = ).
Second argument equals 'Student' ( != false).

```

10. String Zero-Length (-z)

```

if [ -z "$str" ]; then
    echo "Second argument is empty (zero-length)."

```

Commands:

`./setup.sh 10`

`./setup.sh 10 Student`

```

raj@ubuntu-lab:~$ ./setup.sh 10
10 is equal to 10 (-eq).
10 is equal to 10 (-ne false).
10 is NOT greater than 10 (-gt).
10 is NOT less than 10 (-lt).
10 is greater than or equal to 10 (-ge).
10 is less than or equal to 10 (-le).
Second argument does NOT equal 'Student' ( != ).
Second argument is not equal to 'Student' ( != ).
Second argument is empty (zero-length).
raj@ubuntu-lab:~$ ./setup.sh 10 Student
10 is equal to 10 (-eq).
10 is equal to 10 (-ne false).
10 is NOT greater than 10 (-gt).
10 is NOT less than 10 (-lt).
10 is greater than or equal to 10 (-ge).
10 is less than or equal to 10 (-le).
Second argument equals 'Student' ( = ).
Second argument equals 'Student' ( != false).
Second argument is not empty.

```

Task 12: Script setup.sh – Print All Arguments with a For Loop

Goal:

Write a Bash script that prints all user-provided command-line arguments using a for loop with \$*.

1. Shebang and Comment

Code added:

`#!/bin/bash`

`# Script to demonstrate printing all user-entered arguments using $*`

Steps:

1. Open vim:
vim setup.sh
2. Insert the lines above.

```
hajra@ubuntu-lab: ~  
#!/bin/bash  
## Script to demonstrate printing all user-entered arguments using $*  
#
```

3. Save and quit:
:wq
4. Run:
./setup.sh

```
hajra@ubuntu-lab:~$ ./setup.sh  
hajra@ubuntu-lab:~$
```

2. Append the For Loop Using \$*

Steps:

1. Reopen the script in vim:
vim setup.sh
2. Append the code below the comment.

```
# Print all arguments using $*  
echo "Printing all arguments using \${*}:"  
for arg in $*; do  
    echo "Argument: $arg"  
done
```

3. Save and quit (:wq).
4. Make script executable (if not already):
chmod +x setup.sh
5. Run the script with sample arguments:
./setup.sh one "two words" three

```
hajra@ubuntu-lab:~$ chmod +x setup.sh  
hajra@ubuntu-lab:~$ ./setup.sh one "two words" three  
Printing all arguments using $*:  
Argument: one  
Argument: two  
Argument: words  
Argument: three
```

Task 13: Script setup.sh – While Loop Summation and Functions

Goal:

Create a Bash script that:

1. Uses an interactive **while loop** to keep a running total of user-entered numbers until they type q.
2. Defines a **function sum_two()** that performs the same accumulation inside a function.
3. Adds another **function sum_args()** that takes two numeric arguments, adds them, and returns the result.

1. Shebang Line

Steps:

1. Overwrite any existing setup.sh:
vim setup.sh
2. Insert the code line.
3. Save and quit:
:wq

```
hajra@ubuntu-lab: ~
```

```
#!/bin/bash
```

4. Make executable and run (no output expected):

```
chmod +x setup.sh
```

```
./setup.sh
```

```
hajra@ubuntu-lab: $ chmod +x setup.sh
```

```
hajra@ubuntu-lab: $ ./setup.sh
```

2. Interactive While Loop Summation

Steps:

1. Re-open script:
vim setup.sh
2. Append the above.
3. Save and quit (:wq).

```
# While-loop summation (interactive)
sum=0
while true; do
    read -p "Enter a number (or 'q' to quit): " input
    if [ "$input" = "q" ]; then
        break
    fi
    sum=$((sum + input))
    echo "Total Score: $sum"
done
echo "Final total: $sum"
```

4. Run the script and interact:

```
./setup.sh
```

```
hajra@ubuntu-lab: $ ./setup.sh
```

```
Enter a number (or 'q' to quit): 6
```

```
Total Score: 6
```

```
Enter a number (or 'q' to quit): 7
```

```
Total Score: 13
```

```
Enter a number (or 'q' to quit): 8
```

```
Total Score: 21
```

```
Enter a number (or 'q' to quit): 9
```

```
Total Score: 30
```

```
Enter a number (or 'q' to quit): 10
```

```
Total Score: 40
```

```
Enter a number (or 'q' to quit): q
```

```
Final total: 40
```

3. Function sum_two() (Interactive Loop Inside a Function)

Steps:

1. Edit script again and append this function.
2. Save and quit (:wq).

```
# Function to accumulate scores interactively
sum_two() {
    sum=0
    while true; do
        read -p "Enter a number (or 'q' to quit): " input
        if [ "$input" = "q" ]; then
            break
        fi
        sum=$((sum + input))
        echo "Total Score: $sum"
    done
    echo "Function final total: $sum"
}

# Demonstrate the function
echo "Now calling sum_two function:"
sum_two
```

3. Run it:

```
./setup.sh
```

```
Now calling sum_two function:
```

```
Enter a number (or 'q' to quit): 3
```

```
Total Score: 3
```

```
Enter a number (or 'q' to quit): 4
```

```
Total Score: 7
```

```
Enter a number (or 'q' to quit): 5
```

```
Total Score: 12
```

```
Enter a number (or 'q' to quit): q
```

```
Function final total: 12
```

```
hajra@ubuntu-lab:~$
```

4. Function `sum_args()` (Two Arguments Summation)

Steps:

1. Append the code above to `setup.sh`.
2. Save and quit (`:wq`).

```
# Function that sums two arguments and returns the result
sum_args() {
    a=$1
    b=$2
    return $((a + b))
}

# Demonstrate sum_args function
echo "Now demonstrating sum_args function:"
sum_args 3 4
result=$?
echo "sum_args(3,4) returned: $result"
```

3. Run the script:

```
./setup.sh
Now demonstrating sum_args function:
sum_args(3,4) returned: 7
hajra@ubuntu-lab:~$
```

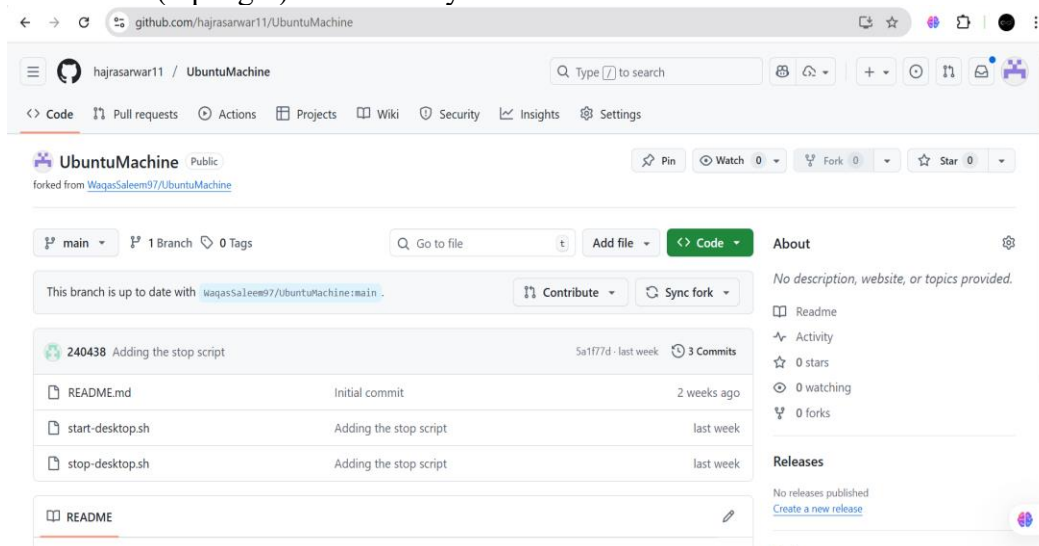
Task 14: Codespaces GUI: Fork Repo, Start GUI, Connect via VNC, Stop GUI

Goal:

Fork a GitHub repository, open it in Codespaces, run the GUI start script, connect via the forwarded port (6080) using `vnc.html`, and stop the GUI.

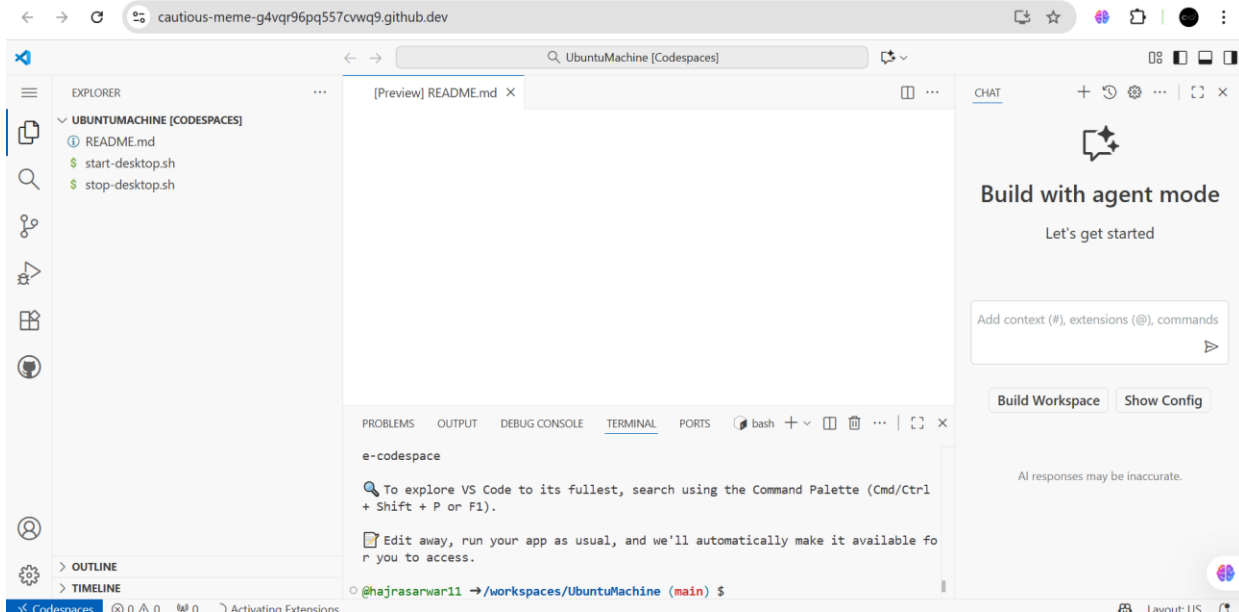
1. Fork the repository to your GitHub account

- Open the repo URL in your browser.
<https://github.com/WaqasSaleem97/UbuntuMachine>
- Click **Fork** (top-right) and fork to your account.



2. Open a Codespace on your fork

- Click **Code** → **Open with Codespaces** → **Create codespace**.
- Wait for Codespace initialization.



3. Verify start/stop scripts exist and are executable

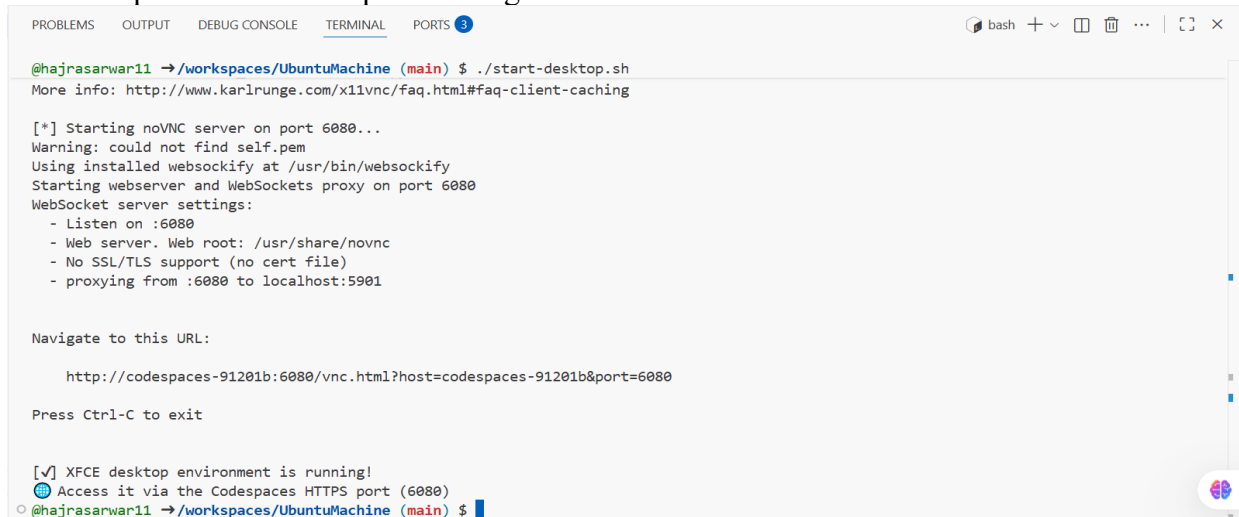
`ls -l start-desktop.sh stop-desktop.sh`

```
@hajrasarwar11 →/workspaces/UbuntuMachine (main) $ ls -l start-desktop.sh stop-desktop.sh
-rwxrwxrwx 1 codespace root 1333 Nov 7 04:43 start-desktop.sh
-rwxrwxrwx 1 codespace root 428 Nov 7 04:43 stop-desktop.sh
@hajrasarwar11 →/workspaces/UbuntuMachine (main) $
```

4. Run the start script to launch the GUI

`./start-desktop.sh`

- Capture terminal output showing successful start.



5. Verify forwarded ports in Codespaces (Ports view)

- Open Ports panel and confirm port 6080 is forwarded.

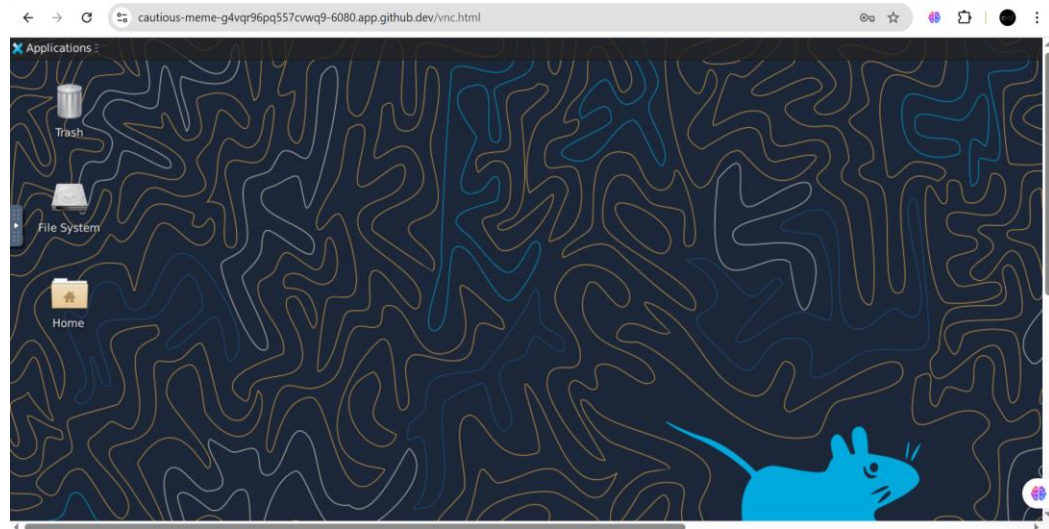
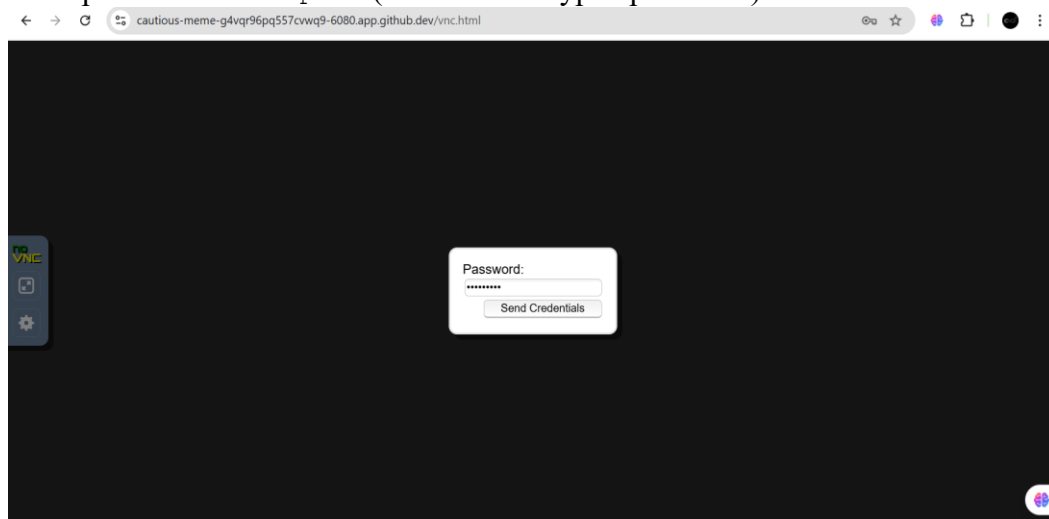
Port	Forwarded Address	Running Process	Visibility	Origin
5900	https://cautious-meme-...	x11vnc -display :1 -rfbauth /home/codespace/vn...	Public	Auto Forwarded
5901	https://cautious-meme-...	x11vnc -display :1 -rfbauth /home/codespace/vn...	Private	Auto Forwarded
6080	https://cautious-meme-...	/usr/bin/python3 /usr/bin/websockify --web /usr...	Public	Auto Forwarded

6. Connect to the VNC HTML page

- Open the forwarded port 6080 in the browser or Codespaces preview.
- Click `vnc.html`.



- Enter password: codespace (do **not** show typed password).



7. Stop the GUI

`./stop-desktop.sh`

- Capture terminal output showing GUI stop/cleanup.


```
hajra@ubuntu-lab:~$ mkdir -p workspace
hajra@ubuntu-lab:~$ touch workspace/secret.txt
hajra@ubuntu-lab:~$ sudo chown examuser:g1 workspace/secret.txt
hajra@ubuntu-lab:~$ ls -l workspace/secret.txt
-rw-rw-r-- 1 examuser g1 0 Nov  7 05:53 workspace/secret.txt
hajra@ubuntu-lab:~$
```

- o **Symbolic:** `chmod go-rwx workspace/secret.txt`
- o **Numeric:** `chmod 700 workspace/secret.txt`

```
hajra@ubuntu-lab:~$ chmod go-rwx workspace/secret.txt
chmod: changing permissions of 'workspace/secret.txt': Operation not permitted
hajra@ubuntu-lab:~$ ls -l workspace/secret.txt
-rw-rw-r-- 1 examuser g1 0 Nov  7 05:53 workspace/secret.txt
hajra@ubuntu-lab:~$ chmod 600 workspace/secret.txt
chmod: changing permissions of 'workspace/secret.txt': Operation not permitted
hajra@ubuntu-lab:~$ ls -l workspace/secret.txt
-rw-rw-r-- 1 examuser g1 0 Nov  7 05:53 workspace/secret.txt
hajra@ubuntu-lab:~$
```

```
ls -l workspace/secret.txt
```

```
hajra@ubuntu-lab:~$ ls -l workspace/secret.txt
-rw-rw-r-- 1 examuser g1 0 Nov  7 05:53 workspace/secret.txt
hajra@ubuntu-lab:~$
```

Scenario: Filter system logs and save results using redirection and piping.

```
grep -E 'error|fail' /var/log/syslog | head -n 20
```

```
root@ubuntu-lab:~# journalctl -n 100 --no-pager --output cat
[{"t": "2025-11-07T03:55:28.576913+00:00", "l": "error/fail", "f": "/var/log/syslog", "h": "head -n 20", "m": "..."}, {"t": "2025-11-07T03:55:28.579319+00:00", "l": "ubuntu-lab microk8s.daemon-containerd[1655]: time=2025-11-07T03:55:28.569735708Z level=info msg=\"skip loading plugin \"io.container.snapshotter.v1.aufs\" ...\", error=aufs is not supported (modprobe aufs failed; exit status 1) modprobe: FATAL: Module aufs not found in directory /lib/modules/6.8.0-86-generic(\\n)\": skip plugin\" type=io.containerd.snapshotter.v1", "m": "..."}, {"t": "2025-11-07T03:55:28.579748+00:00", "l": "ubuntu-lab microk8s.daemon-containerd[1655]: time=2025-11-07T03:55:28.578336862Z level=info msg=\"skip loading plugin \"io.container.snapshotter.v1.btrfs\" ...\", error=path /var/snap/microk8s/common/var/lib/containerd/io.container.snapshotter.v1.btrfs (ext4) must be a btrfs filesystem to be used with the btrfs snapshotter: skip plugin\" type=io.containerd.snapshotter.v1", "m": "..."}, {"t": "2025-11-07T03:55:28.578805+00:00", "l": "ubuntu-lab microk8s.daemon-containerd[1655]: time=2025-11-07T03:55:28.578382472Z level=warning msg=\"failed to load plugin io.container.snapshotter.v1.devmapper\" error=devmapper not configured\"", "m": "..."}, {"t": "2025-11-07T03:55:28.579244+00:00", "l": "ubuntu-lab microk8s.daemon-containerd[1655]: time=2025-11-07T03:55:28.596481822Z level=info msg=\"skip loading plugin \"io.container.snapshotter.v1.zfs\" ...\", error=path /var/snap/microk8s/common/var/lib/containerd/io.container.snapshotter.v1.zfs must be a zfs filesystem to be used with the zfs snapshotter: skip plugin\" type=io.container.snapshotter.v1", "m": "..."}, {"t": "2025-11-07T03:55:28.597448+00:00", "l": "ubuntu-lab microk8s.daemon-containerd[1655]: time=2025-11-07T03:55:28.596575592Z level=warning msg=\"could not use snapshotter devmapper in metadata plugin\" error=devmapper not configured\"", "m": "..."}, {"t": "2025-11-07T03:55:28.624244+00:00", "l": "ubuntu-lab microk8s.daemon-containerd[1655]: time=2025-11-07T03:55:28.620614737Z level=info msg=\"skip loading plugin \"io.container.tracing.processor.v1.otlp\" ...\", error=skip plugin: tracing endpoint not configured\" type=io.container.tracing.processor.v1", "m": "..."}, {"t": "2025-11-07T03:55:28.638896+00:00", "l": "ubuntu-lab microk8s.daemon-containerd[1655]: time=2025-11-07T03:55:28.632148838Z level=info msg=\"skip loading plugin \"io.container.internal.v1.tracing\" ...\", error=skip plugin: tracing endpoint not configured\" type=io.container.internal.v1", "m": "..."}, {"t": "2025-11-07T03:55:28.639215+00:00", "l": "ubuntu-lab microk8s.daemon-containerd[1655]: time=2025-11-07T03:55:28.630649422Z level=info msg=\"Start cri plugin with config {PluginConfig:{ContainerDConfig:{Snapshotter:overlayfs DefaultRuntimeName: default-runtime (Type: Path; Engine: PodAnnotations:[[]] ContainerAnnotations:[[]] Root: Options:map[]) PrivilegedWithoutDevices:false BaseRuntimeSpec: NetworkPluginMaxConfluxNum() Untrusted workload runtime (Type: Path; Engine: PodAnnotations:[[]] ContainerAnnotations:[[]] Root: Options:map[]) PrivilegedWithoutDevices:false BaseRuntimeSpec: NetworkPluginConfDir: NetworkPluginMaxConfluxNum() Runtimes:map[kata:(Type:io.container.kata.v2 Path; Engine: PodAnnotations:[[]] ContainerAnnotations:[[]] Root: Options:map[]) BinaryName:kata-runtime] PrivilegedWithoutDevices:false BaseRuntimeSpec: NetworkPluginConfDir: NetworkPluginMaxConfluxNum() nvidia: container-runtime (Type:io.container.runc.v2 Path; Engine: PodAnnotations:[[]] ContainerAnnotations:[[]] Root: Options:map[]) PrivilegedWithoutDevices:false BaseRuntimeSpec: NetworkPluginConfDir: NetworkPluginMaxConfluxNum() runc:(Type:io.container.runc.v2 Path; Engine: PodAnnotations:[[]] ContainerAnnotations:[[]] Root: Options:map[]) PrivilegedWithoutDevices:false BaseRuntimeSpec: NetworkPluginConfDir: NetworkPluginMaxConfluxNum()) NoPrivilege DisableSnapshots ContainerAnnotations: true DiscardUnpackedLayers: false IgnoreNotFoundErrors:false} cniConfig: {NetworkPluginBinDir:/var/snap/microk8s/8511/opt/cni/bin NetworkPluginConfDir:/var/snap/microk8s/8511/args/cni-network NetworkPluginMaxConfluxNum: 10 NetworkPluginConfTemplate: IPPreference: Ignore (ConfigurePath:/var/snap/microk8s/8511/args/certs.d/0-mirrors.yaml) Configs:map[] Auths:map[] Headers:map[]} ImageEncryption {keyModel:none} DisableCPService:true StreamServerAddress:127.0.0.1 StreamServerPort:2376 StreamServerProtocol:http kcs.io/pause:3.0 StatefulSetPeriod:10 SystemdGroup:false EnableLSTstreaming:false X509CAfile: Streaming:{LSCertFile:LSTKeyFile} MaxContainerLoginSize:16384 DisableGroup:false DisableApparmor:false RestrictOOMScoreAdf:false MaxConcurrentDownloads:3 DisableProcMount:false Unset SeccompProfile: TolerateMissingHubctlController:true DisableHubctlController:true DeviceOwnershipProsecurityContext:false IgnoreUndefinedVolumes:false NETSMToutUnderStateDir:false EnableInPrivilegedGroups:false EnableInPrivilegedCMP:false IgnoreDeprecationWarnings:[[]] DrainK8sSyncTimeout:0s} ContainerRootDir:/var/snap/microk8s/common/var/lib/containerd/containerd binpoint:/var/snap/microk8s/common/run/containerd.sock RootDir:/var/snap/microk8s/common/var/lib/containerd/io.container.grpc.v1.cri StateDir:/var/snap/microk8s/common/run/containerd/io.containerd.io.contain"}", "m": "..."}, {"t": "2025-11-07T03:55:28.650569+00:00", "l": "ubuntu-lab microk8s.daemon-containerd[1655]: time=2025-11-07T03:55:28.637184428Z level=info msg=\"Failed to create contained client\" plugins=idload-image", "m": "..."}, {"t": "2025-11-07T03:55:28.656810+00:00", "l": "ubuntu-lab microk8s.daemon-containerd[1655]: time=2025-11-07T03:55:28.651474692Z level=info msg=\"Failed to create contained client\" plugins=idload-image", "m": "..."}, {"t": "2025-11-07T03:55:28.656964+00:00", "l": "ubuntu-lab microk8s.daemon-containerd[1655]: time=2025-11-07T03:55:28.652029862Z level=info msg=\"Failed to create contained client\" plugins=idload-image", "m": "..."}, {"t": "2025-11-07T03:55:28.723906+00:00", "l": "ubuntu-lab microk8s.daemon-containerd[1655]: time=2025-11-07T03:55:28.714782189Z level=info msg=\"Failed to create contained client\" plugins=idload-image", "m": "..."}, {"t": "2025-11-07T03:55:31.420077+00:00", "l": "ubuntu-lab microk8s.daemon-kubelite[3540]: ++ grep -qE 'Failure domain' /var/snap/microk8s/8511/args/ha-conf", "m": "..."}, {"t": "2025-11-07T03:55:31.448890+00:00", "l": "ubuntu-lab microk8s.daemon-kubelite[3540]: ++ get_opt -n confdir failure_domain ha-conf", "m": "..."}]
```

```
grep -E 'error|fail' /var/log/syslog > ~/logs/errors.txt
grep -E 'warning|fail' /var/log/syslog >> ~/logs/errors.txt
```

```
hajra@ubuntu-lab:~$ mkdir -p ~/logs
hajra@ubuntu-lab:~$ grep -i -E "error|fail" /var/log/syslog > ~/logs/errors.txt
hajra@ubuntu-lab:~$ grep -i -E "warning|fail" /var/log/syslog >> ~/logs/errors.txt
hajra@ubuntu-lab:~$
```

```
less ~/logs/errors.txt
```



```

hajra@ubuntu-lab: ~
workPluginConfDir: NetworkPluginMaxConNum:0} NoPivot:false DisableSnapshotAnnotations:true DiscardUnpackedLayers:false IgnoreRdtNotEnabledErrors:false CniConfig:(NetworkPluginBinDir:/var
/snap/microk8s/8511/opt/cni/bin NetworkPluginConfDir:/var/snap/microk8s/8511/args/cni-network NetworkPluginMaxConNum:1 NetworkPluginConfTemplate: IPPreference:) Registry:(ConfigPath:/var/s
nap/microk8s/8511/args/certs.d Mirrors:map[] Configs:map[] Auths:map[] Headers:map[]) ImageDecryption:{KeyModel:node} DisableTCPSe
2025-11-07T03:55:28.578740:00:00 ubuntu-lab microk8s.daemon-containerd[1655]: time="2025-11-07T03:55:28.578336886Z" level=info msg="skip loading plugin \"io.containerd.snapshotter.v1.btrfs\"
...\" error=\"path /var/snap/microk8s/common/var/lib/containerd/io.containerd.snapshotter.v1.btrfs (ext4) must be a btrfs filesystem to be used with the btrfs snapshotter: skip plugin\" type=
io.containerd.snapshotter.v1
2025-11-07T03:55:28.578805:00:00 ubuntu-lab microk8s.daemon-containerd[1655]: time="2025-11-07T03:55:28.578382477Z" level=warning msg="failed to load plugin io.containerd.snapshotter.v1.dev
mapper\" error=\"devmapper not configured\"
2025-11-07T03:55:28.597385:00:00 ubuntu-lab microk8s.daemon-containerd[1655]: time="2025-11-07T03:55:28.596481842Z" level=info msg="skip loading plugin \"io.containerd.snapshotter.v1.zfs\".
...\" error=\"path /var/snap/microk8s/common/var/lib/containerd/io.containerd.snapshotter.v1.zfs must be a zfs filesystem to be used with the zfs snapshotter: skip plugin\" type=io.containerd.s
napshotter.v1
2025-11-07T03:55:28.597448:00:00 ubuntu-lab microk8s.daemon-containerd[1655]: time="2025-11-07T03:55:28.596557592Z" level=warning msg="could not use snapshotter devmapper in metadata plugin
\" error=\"devmapper not configured\"
2025-11-07T03:55:28.621424:00:00 ubuntu-lab microk8s.daemon-containerd[1655]: time="2025-11-07T03:55:28.620614373Z" level=info msg="skip loading plugin \"io.containerd.tracing.processor.v1.
otlp\". ...\" error=\"skip plugin: tracing endpoint not configured\" type=io.containerd.tracing.processor.v1
2025-11-07T03:55:28.638896:00:00 ubuntu-lab microk8s.daemon-containerd[1655]: time="2025-11-07T03:55:28.632148388Z" level=info msg="skip loading plugin \"io.containerd.internal.v1.tracing\".
...\" error=\"skip plugin: tracing endpoint not configured\" type=io.containerd.internal.v1
2025-11-07T03:55:28.639215:00:00 ubuntu-lab microk8s.daemon-containerd[1655]: time="2025-11-07T03:55:28.636054942Z" level=info msg="Start cri plugin with config {PluginConfig:{ContainerdCon
fig:{Snapshotter:overlayfs DefaultRuntimeName:runc DefaultRuntime:(Type: Path: Engine: PodAnnotations:[ ContainerAnnotations:[ Root: Options:map[] PrivilegedWithoutHostDevices:false BaseR
untimeSpec: NetworkPluginConfDir: NetworkPluginMaxConNum:0} UntrustedWorkloadRuntime:(Type: Path: Engine: PodAnnotations:[ ContainerAnnotations:[ Root: Options:map[] PrivilegedWithoutHost
Devices:false BaseRuntimeSpec: NetworkPluginConfDir: NetworkPluginMaxConNum:0} Runtimes:map[kata:(Type:io.containerd.kata.v2 Path: Engine: PodAnnotations:[ ContainerAnnotations:[ Root:
Options:map[BinaryName:kata-runtime] PrivilegedWithoutHostDevices:false BaseRuntimeSpec: NetworkPluginConfDir: NetworkPluginMaxConNum:0} nvidia-container-runtime:(Type:io.containerd.runc.v
2 Path: Engine: PodAnnotations:[ ContainerAnnotations:[ Root: Options:map[BinaryName:nvidia-container-runtime] PrivilegedWithoutHostDevices:false BaseRuntimeSpec: NetworkPluginConfDir: Ne
tworkPluginMaxConNum:0} runc:(Type:io.containerd.runc.v2 Path: Engine: PodAnnotations:[ ContainerAnnotations:[ Root: Options:map[] PrivilegedWithoutHostDevices:false BaseRuntimeSpec: Net
workPluginConfDir: NetworkPluginMaxConNum:0} NoPivot:false DisableSnapshotAnnotations:true DiscardUnpackedLayers:false IgnoreRdtNotEnabledErrors:false CniConfig:(NetworkPluginBinDir:/var
/snap/microk8s/8511/opt/cni/bin NetworkPluginConfDir:/var/snap/microk8s/8511/args/cni-network NetworkPluginMaxConNum:1 NetworkPluginConfTemplate: IPPreference:) Registry:(ConfigPath:/var/s
nap/microk8s/8511/args/certs.d Mirrors:map[] Configs:map[] Auths:map[] Headers:map[]) ImageDecryption:{KeyModel:node} DisableTCPService:true StreamServerAddress:127.0.0.1 StreamServerPort:0
StreamIdleTimeout:4h0m0s EnableLinux:false SelinuxCategoryRange:1024 SandboxImage:registry.k8s.io/pause:3.10 StatsCollectPeriod:10 SystemdGroup:false EnableLSSStreaming:false X509KeyPair
Streaming:{TLSCertFile: TLSCertFile: MaxContainerLogLineSize:16384 DisableGroup:false DisableApparmor:false RestrictOOMScoreAdj:false MaxConcurrentDownloads:3 DisableProxmount:false Unset
SecompProfile: TolerateMissingGettabController:true DisableIngetabController:true DeviceOwnershipFromSecurityContext:false IgnoreImageDefinedVolumes:false NetNSMountUnderStateDir:false E
nableUnprivilegedPorts:false EnableUnprivilegedICMP:false IgnoreDeprecationWarnings:[ DrainExecSyncIOTimeout:0s] ContainerdRootDir:/var/snap/microk8s/common/var/lib/containerd/containerd
dpoint:/var/snap/microk8s/common/run/containerd.sock RootDir:/var/snap/microk8s/common/var/lib/containerd/io.containerd.grpc.v1.cri StateDir:/var/snap/microk8s/common/run/containerd/io.cont
ainerd.grpc.v1.cri}
2025-11-07T03:55:28.650569:00:00 ubuntu-lab microk8s.daemon-containerd[1655]: time="2025-11-07T03:55:28.637184428Z" level=info msg="Failed to create containerd client\" plugin=sideload-image
s
2025-11-07T03:55:28.656180:00:00 ubuntu-lab microk8s.daemon-containerd[1655]: time="2025-11-07T03:55:28.651471469Z" level=info msg="Failed to create containerd client\" plugin=sideload-image
s
2025-11-07T03:55:28.656964:00:00 ubuntu-lab microk8s.daemon-containerd[1655]: time="2025-11-07T03:55:28.652029826Z" level=info msg="Failed to create containerd client\" plugin=sideload-image
s
2025-11-07T03:55:28.723906:00:00 ubuntu-lab microk8s.daemon-containerd[1655]: time="2025-11-07T03:55:28.714782189Z" level=info msg="Failed to create containerd client\" plugin=sideload-image
s
2025-11-07T03:55:31.420877:00:00 ubuntu-lab microk8s.daemon-kubelite[3540]: ++ grep -qE \"failure-domain\" /var/snap/microk8s/8511/args/ha-conf
2025-11-07T03:55:31.448900:00:00 ubuntu-lab microk8s.daemon-kubelite[3545]: ++ get_opt_in_config failure-domain ha-conf
/home/hajra/logs/errors.txt

```

4. Script: Variables, Command Substitution, File & Dir Checks

Scenario: Build and run a script incrementally demonstrating variables, command substitution, and filesystem checks.

1. Create `setup.sh` with shebang and variable `var1`

```

hajra@ubuntu-lab: ~
#!/bin/bash
var1="Hello Lab 6"
echo "var1: $var1"

hajra@ubuntu-lab: $ vim setup.sh
hajra@ubuntu-lab: $ chmod +x setup.sh
hajra@ubuntu-lab: $ ./setup.sh
var1: Hello Lab 6
hajra@ubuntu-lab: $

```

2. Append command substitution storing `ls -l` output

```

files_list=$(ls -l)
echo "Current directory listing:"
echo "$files_list"

hajra@ubuntu-lab: $ vim setup.sh
hajra@ubuntu-lab:~$ ./setup.sh
var1: Hello Lab 6
Current directory listing:
total 411828
-rw-rw-r-- 1 hajra hajra      385 Oct 23 12:56 answers.md
-rw-rw-r-- 1 hajra hajra      684 Oct 24 20:28 apt_update_vs_upgrade.md
drwxr-xr-x 2 hajra hajra    4096 Oct 24 21:25 Desktop
-rw-rw-r-- 1 hajra hajra 421490404 Oct 23 14:32 docker-desktop-amd64.deb
drwxr-xr-x 2 hajra hajra    4096 Oct 24 21:25 Documents
drwxr-xr-x 2 hajra hajra    4096 Oct 25 17:38 Downloads
drwxrwxr-x 2 hajra hajra    4096 Oct 23 04:41 feswdr
-rw-rw-r-- 1 hajra hajra         0 Nov  4 09:09 journal_errors.txt
drwxrwxr-x 3 hajra hajra    4096 Oct 23 14:31 laab4
drwxrwxr-x 3 hajra hajra    4096 Oct 23 13:11 lab4
drwxrwxr-x 2 hajra hajra    4096 Oct 25 15:36 Lab5
drwxrwxr-x 2 hajra hajra    4096 Nov  7 06:01 logs
drwxr-xr-x 2 hajra hajra    4096 Oct 24 21:25 Music
drwxr-xr-x 2 hajra hajra    4096 Oct 24 21:25 Pictures
drwxr-xr-x 2 hajra hajra    4096 Oct 24 21:25 Public
-rwxrwxr-x 1 hajra hajra      125 Nov  7 06:08 setup.sh
drwx----- 3 hajra hajra    4096 Oct 24 20:44 snap
-rw-rw-r-- 1 hajra hajra 125356 Nov  4 09:06 syslog_systemd.txt
-rw-r--r-- 1 hajra hajra   9839 Sep 26 22:06 systemctl status ssh
drwxr-xr-x 2 hajra hajra    4096 Oct 24 21:25 Templates
drwxrwxr-x 2 hajra hajra    4096 Oct 24 21:25 thinclient_drives
drwxr-xr-x 2 hajra hajra    4096 Oct 24 21:25 Videos
drwxrwxr-x 2 hajra hajra    4096 Nov  7 05:53 workspace
hajra@ubuntu-lab: $

```

3. Append directory/file checks for dir1 and dir1/file2

```
if [ ! -d "dir1" ]; then
    mkdir -p dir1
fi
if [ ! -f "dir1/file2" ]; then
    touch dir1/file2
    chmod 644 dir1/file2
fi
ls -l dir1 dir1/file2

dir1:
total 0
-rw-r--r-- 1 hajra hajra 0 Nov  7 06:11 file2
hajra@ubuntu-lab:~$
```

5. Script: Comparisons and String Tests

Scenario: Incrementally add numeric and string comparison tests and show both true/false cases.

1. Overwrite setup.sh to set num=\$1 and str=\$2; add -eq test

```
hajra@ubuntu-lab:~$ ./setup.sh 5 hello
Testing numeric equality (-eq):
5 is equal to 5 (true)
5 is not equal to 10 (false)
hajra@ubuntu-lab:~$
```

2. Append numeric tests: -ne, -gt, -lt, -ge, -le

```
hajra@ubuntu-lab:~$ vim setup.sh
hajra@ubuntu-lab:~$ ./setup.sh 5 test
Testing numeric equality (-eq):
5 is equal to 5 (true)
5 is not equal to 10 (false)

Other numeric comparison tests:
5 is equal to 5
5 is greater than 3
5 is less than 8
5 is greater than or equal to 5
5 is less than or equal to 5
hajra@ubuntu-lab:~$ ./setup.sh 8 test
Testing numeric equality (-eq):
8 is not equal to 5 (false)
8 is not equal to 10 (false)

Other numeric comparison tests:
8 is not equal to 5
8 is greater than 3
8 is not less than 8
8 is greater than or equal to 5
8 is greater than 5
```

3. Append string tests: equality (=), inequality (!=), zero-length (-z)

```
hajra@ubuntu-lab:~$ vim setup.sh
hajra@ubuntu-lab:~$ ./setup.sh 5 hello
Testing numeric equality (-eq):
5 is equal to 5 (true)
5 is not equal to 10 (false)

Other numeric comparison tests:
5 is equal to 5
5 is greater than 3
5 is less than 8
5 is greater than or equal to 5
5 is less than or equal to 5

String comparison tests:
./setup.sh: line 60: syntax error near unexpected token '('
./setup.sh: line 60: `String equality (-)'
hajra@ubuntu-lab:~$ ./setup.sh 5 world
Testing numeric equality (-eq):
5 is equal to 5 (true)
5 is not equal to 10 (false)

Other numeric comparison tests:
5 is equal to 5
5 is greater than 3
5 is less than 8
5 is greater than or equal to 5
5 is less than or equal to 5

String comparison tests:
./setup.sh: line 60: syntax error near unexpected token '('
./setup.sh: line 60: `String equality (-)'
hajra@ubuntu-lab:~$ ./setup.sh 5 ""
Testing numeric equality (-eq):
5 is equal to 5 (true)
5 is not equal to 10 (false)

Other numeric comparison tests:
5 is equal to 5
5 is greater than 3
5 is less than 8
5 is greater than or equal to 5
5 is less than or equal to 5

String comparison tests:
./setup.sh: line 60: syntax error near unexpected token '('
./setup.sh: line 60: `String equality (-)'
```

6. Script: For Loop and Argument Handling

Scenario: Print all provided arguments and demonstrate correct handling of quoted multi-word arguments.

1. Create/overwrite `setup.sh` and print each argument using a for loop

```
hajra@ubuntu-lab: ~  
#!/bin/bash  
s# Script to print all provided arguments  
  
echo "Printing all arguments:"  
for arg in "$@"  
do  
    echo "$arg"  
done
```

2. Run script with single and quoted multi-word arguments

```
./setup.sh one "two words" three  
hajra@ubuntu-lab: $ chmod +x setup.sh  
hajra@ubuntu-lab: $ ./setup.sh one "two words" three  
./setup.sh: line 2: s#: command not found  
Printing all arguments:  
one  
two words  
three
```

7. Script: While Loop Summation and Functions

Scenario: Implement an interactive summation function and a function returning a numeric result.

1. Interactive while-loop that accumulates numbers until `q`

```
hajra@ubuntu-lab:~$ vim setup.sh  
hajra@ubuntu-lab:~$ ./setup.sh  
Enter a number (or 'q' to quit): 7  
Current total: 7  
Enter a number (or 'q' to quit): 8  
Current total: 15  
Enter a number (or 'q' to quit): 9  
Current total: 24  
Enter a number (or 'q' to quit): q  
Final sum: 24
```

2. Function that sums two numeric arguments and demonstrates capturing the result

```
hajra@ubuntu-lab: $ vim setup.sh  
hajra@ubuntu-lab: $ ./setup.sh  
Enter a number (or 'q' to quit): 7  
Current total: 7  
Enter a number (or 'q' to quit): 8  
Current total: 15  
Enter a number (or 'q' to quit): q  
Final sum: 15  
The sum of 7 and 8 using function is: 15
```
